

Bioinformatics Bootcamp 2026

Assessment 02

PART-1

Covered Lectures:

- Lecture 03: Protein Databases
- Lecture 04: Data Retrieval from Biological Databases (NCBI, UniProt & PDB)

Core Challenge

Modern biological research depends on efficiently retrieving, interpreting, and integrating information from multiple biological databases. Selecting the appropriate database is often the first and most critical step in solving a biological research problem.

Question 1

A molecular biologist has been provided with only a gene accession number from a published research article and needs to retrieve the corresponding DNA sequence, protein sequence, associated literature, and gene information.

Describe the step-by-step approach you would follow using NCBI resources. Explain which databases and tools you would access during this process and why.

Question 2

A structural biologist is interested in studying the three-dimensional structure of the human insulin protein to understand how mutations may affect its function.

Which database would you recommend for this purpose? Explain how you would retrieve the structure and describe the information available within the database that would support the research.

Question 3

Imagine you have been assigned a protein with no prior knowledge except its accession ID.

Design a complete workflow describing how you would retrieve, analyze, and interpret all available information about this protein using NCBI, UniProt, and the Protein Data Bank (PDB). Explain the purpose of each step and how the information from one database complements the others.

Submission Instructions

- Answer all questions in your own words.
- Support your answers with scientific reasoning.
- Where appropriate, explain *why* a particular database is the most suitable choice.
- Avoid copying from AI tools or online sources.
- Recommended length: **150–300 words per question.**
- Submit your work as a **single PDF document.**

PART-2

Sequence Retrieval Exercise

Choose **one** biological sequence (DNA or Protein) from **NCBI, UniProt, or the Protein Data Bank (PDB).**

Include the following in your submission:

- 📄 **Screenshot** of the retrieved database record.
- 📄 **Accession ID** or Entry ID of the selected sequence.
- 📄 **Name of the Gene/Protein.**
- 📄 **Source Organism.**
- 📄 A brief description (**150–250 words**) explaining:
 - What the sequence or protein is.
 - Its biological function.
 - Why it is important.
 - Which database you used and why you selected it.

Submission Requirements

Your PDF should include:

- Answers to all assessment questions.
- The practical sequence retrieval exercise.
- At least **one screenshot** of the retrieved database entry.
- Proper headings for each section.
- Post Your working on your social media platforms.

Important Notes

- **Submission Deadline: Wednesday, 17 July 2026**
- Outstanding submissions will be **featured and nominated for publication** on the official **BioCode Innovators** article page and GitHub repository in recognition of academic excellence.
- Submit your assessment as **one PDF file**.
- Your work should reflect **critical thinking, conceptual understanding, originality, and your own analysis** of the topics covered.
- **Well-reasoned, evidence-based, and original responses are highly encouraged.**
- Plagiarism or direct copying from online sources or AI-generated content should be avoided. Use AI only as a learning aid, not as a replacement for your own reasoning and understanding.